



Artificial Intelligence and Machine Learning

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Department of Computer Applications

Artificial Intelligence and Machine Learning

AI Landscape

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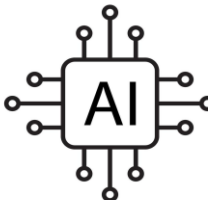
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What is Rationality?

Logical Decision-Making: Following rules to achieve desirable outcomes.

Maximizing Benefit: Actions should yield the maximum benefit.

Goal-Oriented: Decisions are aimed at achieving specific goals.



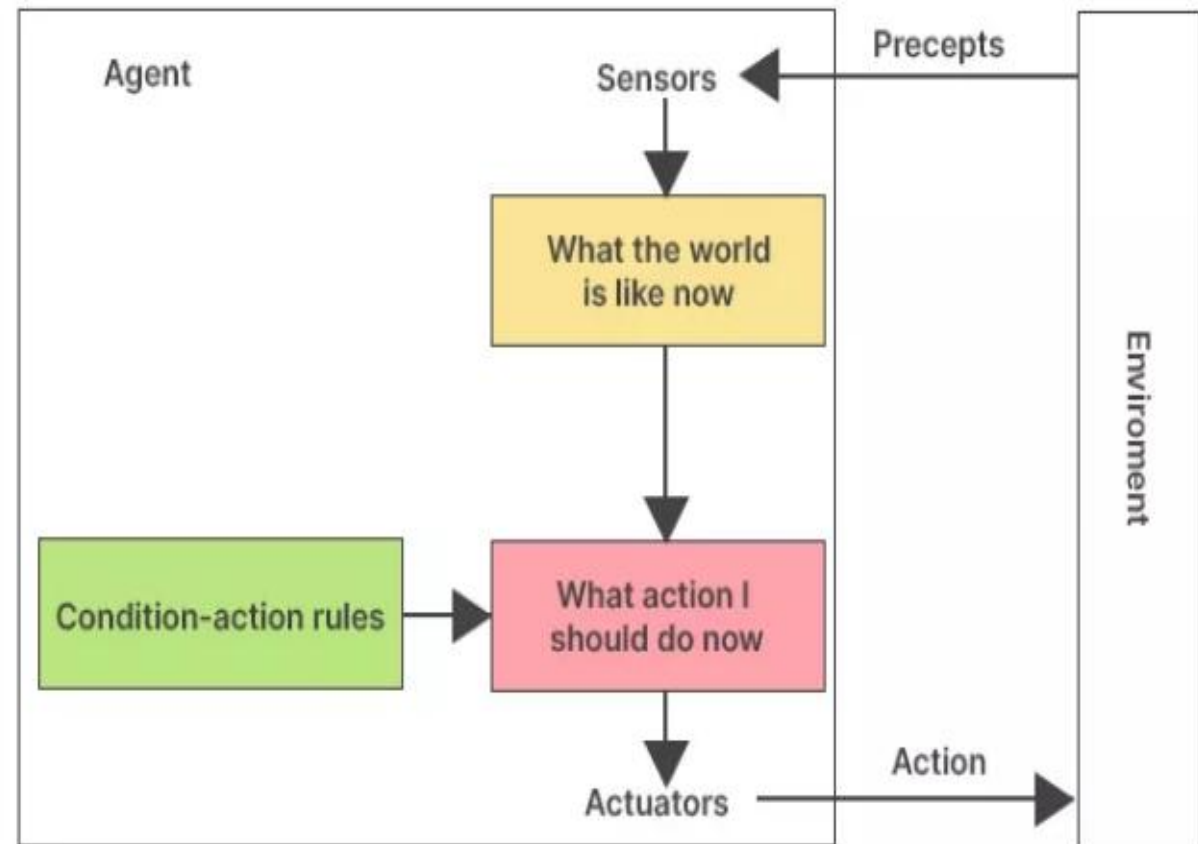
Recap: What is an Agent?

Entities that perceive and act upon their environment.

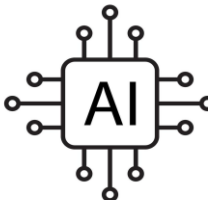
Types:

Software Agents: Interact with digital environments.

Robotic Agents: Operate in the physical world.



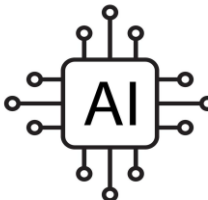
An agent has to act rationally if, given set of rules, takes actions to achieve goals



What is a Rational Agent?

A Rational Agent is an agent that chooses actions that maximize its performance measure based on its percept sequence and knowledge.

A rational agent always does the **best possible action** based on what it knows.



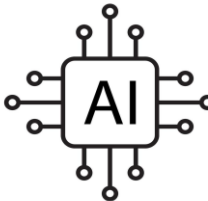
Key Idea Behind Rationality

Rational = Doing the right thing

But...

Right thing according to:

- Available information
- Goals
- Performance measure



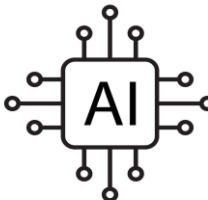
Rationality $\neq$ Omniscient

Omniscient Agent:

- Knows everything
- Knows future outcomes

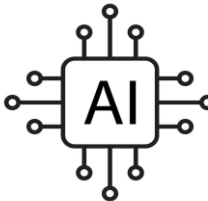
Rational Agent:

- Uses available information
- Makes best possible decision
- Rationality does NOT require perfection



Why do we need Rational Agents?

- Agents must choose actions
- How to decide which action is best?
- That leads to rationality



Rational Agent Example1: Student Writing an Exam

Percepts:

Question paper

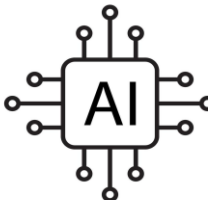
Time available

Action:

Choosing which question to answer first

Rational action:

Attempt easy questions first to maximize marks



Rational Agent Example2: Self-Driving Car

Percepts:

Road Condition

Traffic Signals

Speed of other Vehicles

Action:

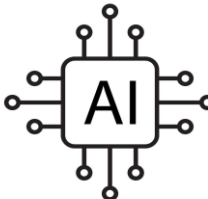
Safety

Comfort

Fuel Efficiency

Rational action:

Slowing down in traffic





THANK YOU

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